

Research Project: Literature Review & Presentation

Intro to Computer Vision

Spring 2026

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1 Project Overview

The goal of this project is to conduct a focused literature review on a research topic of interest, synthesize recent and classical papers, and present your findings both in written and oral form. You will work in groups of 1–3 students and deliver a short presentation (≥ 5 minutes per person) and a 2-page LaTeX report.

2 Project Goals

By the end of this project, you should be able to:

- Identify and read 2–3 key research papers in a focused area
- Synthesize complex technical material into accessible summaries
- Distinguish between essential concepts and secondary details
- Communicate your findings clearly to peers (oral presentation)
- Document your work professionally using LaTeX
- (Optional) Implement a working model or proof-of-concept to deepen understanding

3 Key Requirements

3.1 Literature Scope

- Select and read 2–3 core papers on your chosen topic
- Papers may come from the suggested list, or you can propose your own (mark as [new] in submissions)
- Overlap between groups is acceptable but minimize repeated papers where possible
- Look for both classical and recent work where applicable

3.2 Group Composition

- Groups of 1, 2, or 3 students
- Every group member must speak during the presentation
- Minimum speaking time: 5 minutes per person
- All students must be involved in paper selection, research, and preparation

3.3 Mandatory Participation

- Every person presents—no exceptions
- Share speaking time evenly (roughly 5 min/person for a 2-person group = 10 min total, etc.)
- Coordinate internally to divide content logically (e.g., by paper, by topic section, by methodology)

3.4 Custom Papers

- Proposing papers not on the suggested list? Use **[new]** notation in the Excel submission form
- Instructor will review and approve/reject within a reasonable timeframe
- Ensure your proposed papers are relevant, peer-reviewed, and of appropriate scope

3.5 Paper Tracking & Coordination

- An Excel file will be shared (after Monday) for groups to register their chosen papers
- Use this sheet to:
 - List your group members
 - Declare your chosen papers (2–3 titles)
 - Mark **[new]** papers for instructor approval
- Check this sheet regularly to avoid excessive overlap with other groups

4 Deliverables

4.1 1. Oral Presentation (20 points)

Duration:

- 5 minutes minimum per person
- Total: $5 \text{ min} \times \text{number of group members}$
- Example: 2-person group = 10 min total

Content:

- Introduce your research question/topic
- Summarize the 2–3 core papers clearly and concisely
- Explain key concepts, methods, or findings
- Highlight connections between papers
- Include motivation (why this topic matters)
- Conclude with implications or open questions

Format:

- Slides recommended (PowerPoint, Beamer, Google Slides, etc.)
- Live demo or visualization strongly encouraged if applicable

4.2 2. Written Report (20 points)

Format:

- 2 pages (maximum)
- LaTeX required (provide `.tex` file and compiled PDF)
- Single-spaced, standard margins, 11–12pt font

Content:

- Introduction: motivate the topic and define the research question
- Main body: synthesize the 2–3 papers
 - Describe classical and recent contributions
 - Explain methodologies, key results, and limitations
 - Highlight connections and differences between papers
- Conclusion: summarize takeaways and suggest future directions
- References: cite all papers (BibTeX style preferred)

4.3 3. Optional: Working Model / Proof-of-Concept (5–10% bonus)

If applicable to your topic, implement:

- A small-scale experiment or reproduction of a key result
- A simple demonstration or visualization
- Code in any language (Python, MATLAB, etc.)
- Recommended: use Google Colab for easy sharing and reproducibility

Bonus applies if:

- The model works and produces reasonable results
- Code is well-documented and reproducible
- You discuss how it validates or extends the papers' findings
- Note: This is optional but can help if presentation/report grades are borderline

5 Grading Breakdown

Component	Points
Presentation	20
Report	20
Total	40
<i>+ up to 10% bonus for working model</i>	

Criteria:

- Both presentation and report are graded on clarity, understanding, organization, and accuracy
- Each group member receives the same presentation grade
- If a group member does not speak, they will receive a reduced grade
- The report grade is shared, but each member must sign off (acknowledge contribution)

6 Important Notes & FAQs

6.1 How do I choose a topic?

1. Review the suggested paper list provided (see Appendix A)
2. Consult related topics to see if there's overlap (coordinate early)
3. If you have a unique idea, propose it with a `[new]` marker
4. If stuck, prioritize earlier topics in the list (they're more foundational)
5. Ask the instructor for guidance if uncertain

6.2 What if our papers are similar to another group's?

- This is acceptable—papers may be cited across multiple topics
- However, try to minimize direct overlap by coordinating in the Excel sheet
- If unavoidable, explain in your presentation how your angle differs

6.3 Can I propose papers not on the suggested list?

- Yes! Use the `[new]` notation in the Excel submission form

6.4 Can we include experiments or code in the presentation?

- Yes! This is highly encouraged
- Live demos, visualizations, and code walkthroughs are engaging
- If showing code, keep it brief and well-commented

6.5 How do I format the LaTeX report?

- Use standard LaTeX document class (`article`)
- Include `\documentclass{article}` with geometry for margins
- Use BibTeX for references
- Compile to PDF before submission
- Ensure figures and citations are properly formatted
- See template provided (or ask instructor for example)

6.6 What counts as a 'working model'?

- Reproduction of a key result from one of your papers
- A simple implementation of a method described in the literature
- A visualization or interactive demo illustrating core concepts
- Must be runnable and produce meaningful output
- 5–10 lines of code or a small-scale experiment is sufficient

6.7 How do I submit my work?

- Presentation: Submit slides before your scheduled time
- Report: Upload pdf to brightspace
- Code (optional): Upload GitHub URL in your report
- All submissions must be clearly labeled with group members' names and date

7 Submission Instructions

Please prepare and submit the following:

7.1 1. Presentation Slides

- Format: PDF, PowerPoint, or Beamer output
- File name: `GroupName_Presentation.pdf` (or `.pptx`)
- Submit by: Day before your presentation slot

7.2 2. LaTeX Report (Required)

- File names: `GroupName_Report.tex` and `GroupName_Report.pdf`
- Check that LaTeX compiles without errors
- Page limit: 2 pages (strictly enforced)
- Submit by: Same day as presentation

7.3 3. Model / Code (Optional, for bonus)

- Language: Python, MATLAB, or other
- Format: `.zip` file or GitHub link with README
- Include instructions to run the code
- File name: `GroupName_Code.zip`
- Submit by: Same day as presentation

A Research Topics, Theory & Suggested Papers

A.1 Image Denoising, Deblurring, Superresolution

Theory:

Image restoration is a fundamental problem in computer vision: recovering a clean, sharp image from a degraded observation. Degradation can take many forms: additive noise (random pixels), motion blur (camera movement), defocus blur (poor focus), or downsampling (loss of resolution). The goal is to invert or undo these degradations.

Key challenges:

- Ill-posed problem: many clean images could produce the same degraded observation
- Balancing fidelity (staying close to input) vs. smoothness/naturalness
- Computational efficiency for real-time applications

Classical approaches rely on optimization: define a cost function (data fidelity + regularization) and minimize it. Modern deep learning approaches learn to predict clean images directly from noisy inputs or use generative models (diffusion) to sample from the distribution of clean images.

Applications: medical imaging, surveillance, photography, video enhancement.

Key Papers:

- Levin, A., Weiss, Y., Durand, F., & Freeman, W. T. “Understanding and evaluating blind deconvolution algorithms.” CVPR 2009
- Mairal, J., Bach, F., Ponce, J., Sapiro, G., & Zisserman, A. “Non-local sparse models for image restoration.” ICCV 2009
- Ulyanov, D., Vedaldi, A., & Lempitsky, V. “Deep image prior.” CVPR 2018
- Glasner, D., Bagon, S., & Irani, M. “Super-resolution from a single image.” ICCV 2009
- Michaeli, T., & Irani, M. “Blind deblurring using internal patch recurrence.” ECCV 2014
- Shocher, A., Cohen, N., & Irani, M. “Zero-shot super-resolution using deep internal learning.” CVPR 2018
- Zoran, D., & Weiss, Y. “From learning models of natural image patches to whole image restoration.” ICCV 2011
- Whyte, O., Sivic, J., Zisserman, A., & Ponce, J. “Non-uniform deblurring for shaken images.” IJCV 2012
- Kavar, B., Elad, M., Ermon, S., & Song, J. “Denoising Diffusion Restoration Model.” NeurIPS 2022

A.2 Scattering Transform

Theory:

The scattering transform is a mathematical framework that provides a theoretically grounded explanation for why Convolutional Neural Networks (CNNs) work so well. Developed by Stéphane Mallat, it builds a multi-scale representation by iteratively applying wavelet transforms and nonlinearities.

Key ideas:

- Wavelets capture image patterns at multiple scales and orientations

- Cascading (iterating) wavelet transforms + absolute value (nonlinearity) creates invariance to small translations and stability to small deformations
- The resulting feature representation is surprisingly similar to what CNNs learn
- Provides theoretical guarantees on stability and approximation properties

Advantages: theoretically justified, interpretable (wavelets have known meaning), requires no training.

Disadvantages: less flexible than learned representations, may be outperformed by end-to-end deep learning on large datasets.

Warning: This topic requires comfort with mathematics (functional analysis, harmonic analysis). It offers deep insights into the theoretical foundations of deep learning.

Key Papers:

- Mallat, S. “Understanding deep convolutional networks.” Phil. Trans. R. Soc. 2016
- Sifre, L., & Mallat, S. “Rotation, scaling and deformation invariant scattering for texture discrimination.” CVPR 2013
- Oyallon, E., & Mallat, S. “Deep roto-translation scattering for object classification.” CVPR 2015
- Sifre, L. “Rigid-motion scattering for image classification.” Ph.D. dissertation 2014
- Oyallon, E., Belilovsky, E., & Zagoruyko, S. “Scaling the scattering transform: Deep hybrid networks.” ICCV 2017

A.3 Local Image Detectors and Descriptors

Theory:

Local features are keypoints (distinctive locations) and descriptors (representations of their neighborhoods) that are invariant to transformations like rotation, scaling, and affine distortions. They form the foundation of classical computer vision: matching features across images enables 3D reconstruction, object recognition, and image alignment.

Core concepts:

- **Detection:** Finding distinctive points (corners, blobs, edges) where image structure is rich
- **Description:** Encoding the neighborhood around each point in a compact, robust way
- **Invariance:** Handling rotation, scale, and affine transformations
- **Distinctiveness:** Different points should have very different descriptors

Classical methods (SIFT, SURF) relied on hand-crafted features. Recent work (LIFT, SuperPoint, D2-Net) learns features from data.

This is a foundational topic for understanding feature-based pipelines and the transition from classical to learning-based methods.

Key Papers:

- Mikolajczyk, K., & Schmid, C. “Scale & affine invariant interest point detectors.” IJCV 2004
- Schmid, C., & Mikolajczyk, K. “A performance evaluation of local descriptors.” ICPR 2005

- Bay, H., Tuytelaars, T., & Van Gool, L. “SURF: Speeded up robust features.” ECCV 2006
- Yi, K. M., Trulls, E., Lepetit, V., & Fua, P. “LIFT: Learned invariant feature transform.” ECCV 2016
- DeTone, D., Malisiewicz, T., & Rabinovich, A. “SuperPoint: Self-supervised interest point detection and description.” CVPR 2018
- Balntas, V., Lenc, K., Vedaldi, A., & Mikolajczyk, K. “HPatches: A benchmark and evaluation of handcrafted and learned local descriptors.” CVPR 2017
- Luo, Z., Shen, T., Zhou, L., Zhang, J., Yao, Y., Li, S., . . . , Quan, L. “ContextDesc: Local Descriptor Augmentation with Cross-Modality Context.” CVPR 2019
- Dusmanu, M., Rocco, I., Pajdla, T., Pollefeys, M., Sivic, J., Torii, A., & Sattler, T. “D2-Net: A Trainable CNN for Joint Detection and Description of Local Features.” arXiv 2019

A.4 Classical Object Detection

Theory:

Object detection answers: “Where are the objects of interest in this image?” It combines localization (bounding boxes) with classification (what is the object?).

Classical methods (1990s–2010s):

- **Sliding window:** Apply a classifier at every position and scale (computationally expensive)
- **Hand-crafted features:** Combine HOG, SIFT, bag-of-words with classifiers (Boosted cascades, SVM, DPM)
- **Selective search:** Use bottom-up segmentation to propose regions, then classify

Deep learning methods (2010s–present):

- **Region-based:** Faster R-CNN, Mask R-CNN (propose regions, then refine)
- **Single-shot:** YOLO, SSD (directly predict boxes and classes)
- **Transformer-based:** DETR (treat as set prediction problem)

Key challenges: speed/accuracy trade-off, handling multiple scales, class imbalance, small objects.

Key Papers:

- Viola, P., & Jones, M. “Rapid object detection using a boosted cascade of simple features.” CVPR 2001
- Dalal, N., & Triggs, B. “Histograms of oriented gradients for human detection.” CVPR 2005
- Hariharan, B., Malik, J., & Ramanan, D. “Discriminative decorrelation for clustering and classification.” ECCV 2012
- Felzenszwalb, P., Girshick, R., McAllester, D., & Ramanan, D. “Object detection with discriminatively trained part-based models.” PAMI 2010

- Philbin, J., Chum, O., Isard, M., Sivic, J., & Zisserman, A. “Object retrieval with large vocabularies and fast spatial matching.” CVPR 2007
- *Note: RCNN, Fast RCNN, Faster-RCNN, Mask RCNN covered in lecture; if included, provide additional depth*
- Redmon, J., Divvala, S., Girshick, R., et al. “You only look once: Unified, real-time object detection.” CVPR 2016
- Carion, N., Massa, F., Synnaeve, G., et al. “End-to-end object detection with transformers.” ECCV 2020

A.5 Aggregating Features for Recognition

Theory:

Image recognition requires converting a set of local features (e.g., hundreds of SIFT descriptors) into a single compact representation for classification. This is the “pooling” or “aggregation” problem.

Key approaches:

- **Bag of Words:** Quantize features into a visual vocabulary, count occurrences
- **Spatial Pyramid Matching:** Divide image into grids, compute bag-of-words per grid, combine hierarchically
- **VLAD (Vector of Locally Aggregated Descriptors):** Assign features to nearest vocabulary words, sum residuals
- **Fisher Encoding:** Model features with GMM, encode differences from GMM parameters
- **NetVLAD:** Learn aggregation end-to-end in a neural network

These methods bridge local features (descriptive but high-dimensional) and global image labels (interpretable but loss of information).

Key Papers:

- Lazebnik, S., Schmid, C., & Ponce, J. “Beyond bags of features: Spatial pyramid matching for recognizing natural scene categories.” CVPR 2006
- Jégou, H., Douze, M., Schmid, C., & Pérez, P. “Aggregating local descriptors into a compact image representation.” CVPR 2010
- Perronnin, F., Sánchez, J., & Mensink, T. “Improving the Fisher kernel for large-scale image classification.” ECCV 2010
- Arandjelovic, R., Gronat, P., Torii, A., Pajdla, T., & Sivic, J. “NetVLAD: CNN architecture for weakly supervised place recognition.” CVPR 2016

A.6 Semantic Segmentation

Theory:

Semantic segmentation assigns a class label to every pixel in an image. Unlike object detection (bounding boxes), it provides pixel-level precision. Applications: medical imaging, autonomous driving, scene understanding.

Key challenges:

- Preserving fine spatial details while capturing context (encoder-decoder architecture)

- Handling objects at multiple scales
- Class imbalance (some classes much rarer than others)
- Computational cost (processing every pixel)

Core architectures:

- **FCN (Fully Convolutional Networks):** Replace fully connected layers with convolutions; upsampling via deconvolution
- **U-Net:** Encoder-decoder with skip connections; popular in medical imaging
- **DeepLab:** Atrous convolution for multi-scale receptive fields + CRF for refinement
- **Transformer-based:** Attend to global context; recent state-of-the-art

Recent advances include attention mechanisms (CCNet), neural architecture search (Auto-DeepLab), and foundation models (SAM).

Note: The first papers' main ideas are covered in lecture. Focus on novel aspects and recent progress.

Key Papers:

- Long, J., Shelhamer, E., & Darrell, T. "Fully convolutional networks for semantic segmentation." CVPR 2015
- Ronneberger, O., Fischer, P., & Brox, T. "U-Net: Convolutional networks for biomedical image segmentation." MICCAI 2015
- Hariharan, B., Arbeláez, P., Girshick, R., & Malik, J. "Hypercolumns for object segmentation and fine-grained localization." CVPR 2015
- He, K., Gkioxari, G., Dollár, P., & Girshick, R. "Mask R-CNN." ICCV 2017 (*note: also in detection project*)
- Chen, L. C., Papandreou, G., Kokkinos, I., Murphy, K., & Yuille, A. L. "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs." PAMI 2017
- Jégou, S., Drozdal, M., Vazquez, D., Romero, A., & Bengio, Y. "The one hundred layers tiramisu: Fully convolutional densenets for semantic segmentation." CVPR Workshops 2017
- Chen, L. C., Zhu, Y., Papandreou, G., Schroff, F., & Adam, H. "Encoder-decoder with atrous separable convolution for semantic image segmentation." ECCV 2018
- Liu, C., Chen, L. C., Schroff, F., Adam, H., Hua, W., Yuille, A. L., & Fei-Fei, L. "Auto-DeepLab: Hierarchical neural architecture search for semantic image segmentation." CVPR 2019
- Huang, Z., Wang, X., Huang, L., Huang, C., Wei, Y., & Liu, W. "CCNet: Criss-cross attention for semantic segmentation." ICCV 2019
- LaLonde, R., & Bagci, U. "Capsules for object segmentation." arXiv 2018
- Cheng, B., Schwing, A., & Kirillov, A. "Per-pixel classification is not all you need for semantic segmentation." NeurIPS 2021
- **BONUS recent SOTA:** Kirillov, A., Mintun, E., Ravi, N., et al. "Segment Anything." arXiv 2023

A.7 Few-Shot Recognition

Theory:

Few-shot learning addresses: “How can a model recognize new classes from only a few examples?” This is inspired by human learning—we can identify a new animal from a single photo.

Core approaches:

- **Metric Learning:** Learn an embedding space where examples of the same class are close (Prototypical Networks, Matching Networks, Relation Networks)
- **Meta-Learning (“Learning to Learn”):** Optimize a model’s initialization or learning algorithm for fast adaptation (MAML, Optimization as Meta-Learning)
- **Transductive:** Use unlabeled test data to refine predictions

Key challenge: avoiding overfitting to the few available examples while leveraging prior knowledge from other classes.

Applications: medical diagnosis (rare diseases), new product recognition, robotics, drug discovery.

Key Papers:

- Chen, W. Y., Liu, Y. C., Kira, Z., Wang, Y. C. F., & Huang, J. B. “A closer look at few-shot classification.” ICLR 2019
- Vinyals, O., Blundell, C., Lillicrap, T., & Wierstra, D. “Matching networks for one shot learning.” NIPS 2016
- Gidaris, S., & Komodakis, N. “Dynamic few-shot visual learning without forgetting.” CVPR 2018
- Snell, J., Swersky, K., & Zemel, R. “Prototypical networks for few-shot learning.” NIPS 2017
- Ravi, S., & Larochelle, H. “Optimization as a model for few-shot learning.” NIPS 2016
- Sung, F., Yang, Y., Zhang, L., Xiang, T., Torr, P. H., & Hospedales, T. M. “Learning to compare: Relation network for few-shot learning.” CVPR 2018

A.8 Self-Supervised Feature Learning

Theory:

Self-supervised learning (SSL) learns representations *without* human-labeled data. Instead, it uses signals from the data itself (e.g., different views of the same image, context, temporal order).

Key ideas:

- **Contrastive Learning:** Pull together representations of related inputs, push apart unrelated ones (MoCo, SimCLR, BYOL, SwAV)
- **Predictive Learning:** Predict masked or missing parts (Context Encoding, Colorization, Rotation Prediction)
- **Clustering-based:** Group similar images and learn cluster assignments (DeepCluster)

Why it works: learning useful features requires capturing semantic structure, which emerges from solving self-supervised tasks.

This is a very active area. Recent models (DINO, MAE) achieve remarkable performance on downstream tasks with minimal fine-tuning.

Key Papers:

- Doersch, C., Gupta, A., & Efros, A. A. “Unsupervised visual representation learning by context prediction.” ICCV 2015
- Caron, M., Bojanowski, P., Joulin, A., & Douze, M. “Deep clustering for unsupervised learning of visual features.” ECCV 2018
- Hénaff, O. J., Razavi, A., Doersch, C., Eslami, S. M., & van den Oord, A. “Data-efficient image recognition with contrastive predictive coding.” arXiv 2019
- He, K., Fan, H., Wu, Y., Xie, S., & Girshick, R. “Momentum Contrast for Unsupervised Visual Representation Learning.” arXiv 2019
- Gidaris, S., Singh, P., & Komodakis, N. “Unsupervised representation learning by predicting image rotations.” ICLR 2018
- Pathak, D., Krahenbuhl, P., Donahue, J., Darrell, T., & Efros, A. A. “Context encoders: Feature learning by inpainting.” CVPR 2016
- Zhang, R., Isola, P., & Efros, A. A. “Colorful image colorization.” ECCV 2016
- Agrawal, P., Carreira, J., & Malik, J. “Learning to see by moving.” CVPR 2015
- Bojanowski, P., & Joulin, A. “Unsupervised Learning by Predicting Noise.” ICML 2017
- **BONUS:** Caron, M., Touvron, H., Mishra, I., Jégou, H., Mairal, J., Bojanowski, P., & Joulin, A. “Emerging Properties in Self-Supervised Vision Transformers.” ICCV 2021

A.9 Diffusion Models

Theory:

Diffusion models are generative models that learn to “denoise” images corrupted by progressively added Gaussian noise. By reversing this process, they can generate new images from pure noise.

Process:

- **Forward (Noise) Process:** Gradually add noise to a real image over T steps until it becomes pure noise
- **Reverse (Denoising) Process:** Learn to predict and remove noise at each step; sampling iterates this reverse process

Advantages over GANs:

- More stable training (no adversarial min-max game)
- Better mode coverage (less mode collapse)
- Flexible conditioning (text, images, class labels)

Recent advances: latent diffusion (compress images, diffuse in latent space for efficiency), guidance (control generation with text or images), video generation.

Note: Topics 9 (Diffusion) and 10 (GANs) overlap significantly. Coordinate with GANs group to avoid redundancy.

Key Papers:

- Ho, J., Jain, A., & Abbeel, P. “Denoising diffusion probabilistic models.” NeurIPS 2020
- Dhariwal, P., & Nichol, A. “Diffusion models beat GANs on image synthesis.” NeurIPS 2021
- Ho, J., & Salimans, T. “Classifier-free diffusion guidance.” arXiv 2022
- Rombach, R., Blattmann, A., Lorenz, D., et al. “High-resolution image synthesis with latent diffusion models.” CVPR 2022
- **BONUS:** Zhang, L., Rao, A., & Agrawala, M. “Adding conditional control to text-to-image diffusion models.” ICCV 2023
- **BONUS:** Girdhar, R., Singh, M., Brown, A., et al. “Emu Video: Factorizing Text-to-Video Generation by Explicit Image Conditioning.” arXiv 2023
- *Note: Feel free to check recent SOTA (MovieGen, SORA, etc.) for cultural knowledge*

A.10 Generative Adversarial Networks (GANs)

Theory:

GANs train two networks in a min-max game: a *generator* G that creates fake images, and a *discriminator* D that distinguishes real from fake. They compete until D can no longer tell them apart.

Core motivation: adversarial training encourages G to learn the true data distribution without explicitly defining a loss function.

Key improvements:

- **Wasserstein GANs:** Use Wasserstein distance instead of JS divergence for better gradients
- **Spectral Normalization:** Stabilize training by normalizing discriminator weights
- **Progressive Growing:** Train on low-res images first, gradually add layers for high-res
- **StyleGAN:** Decouple coarse/fine details by injecting style at different layers

Challenges: training instability, mode collapse, evaluating quality (no ground truth).

Applications: image synthesis, style transfer, data augmentation, super-resolution.

Note: Topics 9 (Diffusion) and 10 (GANs) overlap. Coordinate to avoid redundancy.

Key Papers:

- Goodfellow, I. J., Pouget-Abadie, J., Mirza, M., et al. “Generative adversarial nets.” NIPS 2014
- Mao, X., Li, Q., Xie, H., Lau, R. Y., Wang, Z., & Smolley, S. P. “Least squares generative adversarial networks.” arXiv 2016
- Arjovsky, M., Chintala, S., & Bottou, L. “Wasserstein GAN.” arXiv 2017 (*main technical paper—discuss in detail*)
- Gulrajani, I., Ahmed, F., Arjovsky, M., Dumoulin, V., & Courville, A. C. “Improved training of Wasserstein GANs.” NeurIPS 2017
- Miyato, T., Kataoka, T., Koyama, M., & Yoshida, Y. “Spectral normalization for generative adversarial networks.” arXiv 2018

- Karras, T., Aila, T., Laine, S., & Lehtinen, J. “Progressive growing of GANs for improved quality, stability, and variation.” arXiv 2017
- Wang, T. C., Liu, M. Y., Zhu, J. Y., Tao, A., Kautz, J., & Catanzaro, B. “High-resolution image synthesis and semantic manipulation with conditional GANs.” CVPR 2018
- Brock, A., Donahue, J., & Simonyan, K. “Large scale GAN training for high fidelity natural image synthesis.” arXiv 2018
- Karras, T., Laine, S., & Aila, T. “A style-based generator architecture for generative adversarial networks.” CVPR 2019
- **BONUS:** <https://this-person-does-not-exist.com/en>

A.11 Novel View Synthesis, Neural Radiance Fields (NeRFs)

Theory:

Given a set of 2D images of a scene from different viewpoints, can we synthesize novel views (images from previously unseen viewpoints)? This requires understanding 3D geometry and appearance.

Neural Radiance Fields (NeRFs) represent a scene as a continuous function $\mathbf{F} : (\mathbf{x}, \mathbf{d}) \rightarrow (r, g, b, \sigma)$ that maps a 3D location $\mathbf{x}$ and viewing direction $\mathbf{d}$ to color and opacity. Rendering is done via volume rendering (ray marching).

Key ideas:

- Implicit scene representation (function rather than mesh)
- Differentiable rendering enables end-to-end optimization
- Positional encoding handles high-frequency details
- Can optimize from RGB images alone (self-supervised)

Recent improvements: faster training (hash tables, distillation), better quality (depth supervision), dynamic scenes, video.

Applications: VFX, immersive media, 3D reconstruction, view interpolation.

Key Papers:

- Mildenhall, B., Srinivasan, P. P., Tancik, M., Barron, J. T., Ramamoorthi, R., & Ng, R. “NeRF: Representing scenes as neural radiance fields for view synthesis.” Communications of the ACM 2021
- Deng, K., Liu, A., Zhu, J. Y., et al. “Depth-supervised NeRF: Fewer views and faster training for free.” CVPR 2022
- Wang, P., Liu, L., Liu, Y., et al. “Learning Neural Implicit Surfaces by Volume Rendering for Multi-view Reconstruction (NeuS).” NeurIPS 2021
- Müller, T., Evans, A., Schied, C., et al. “Instant neural graphics primitives with a multi-resolution hash encoding.” TOG 2022
- Kerbl, B., Kopanas, G., Leimkühler, T., et al. “3D Gaussian Splatting for Real-Time Radiance Field Rendering.” TOG 2023

A.12 Documents and Text

Theory:

Extracting information from document images involves: detecting and recognizing text, understanding layout, and parsing structure. This bridges computer vision and NLP.

Pipeline:

- **Text Detection:** Localize text regions (boxes or polygons)
- **Text Recognition:** Convert image pixels to characters (OCR)
- **Layout Analysis:** Understand document structure (sections, tables, headers)
- **Information Extraction:** Parse semantic content

Challenges:

- Irregular text (curved, rotated, distorted)
- Multiple languages and scripts
- Scanned vs. digital documents
- Handwriting variability

Modern approaches: end-to-end CNNs for recognition, sequence models (RNNs, Transformers) for character sequences, multimodal models combining vision and language.

Key Papers:

- Yang, X., Yumer, E., Asente, P., Kraley, M., Kifer, D., & Giles, C. L. “Learning to Extract Semantic Structure from Documents Using Multimodal Fully Convolutional Neural Networks.” CVPR 2017
- Graves, A., Fernández, S., Gomez, F., & Schmidhuber, J. “Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks.” ICML 2006
- Graves, A., & Schmidhuber, J. “Offline handwriting recognition with multidimensional recurrent neural networks.” NIPS 2009
- Sutskever, I., Vinyals, O., & Le, Q. V. “Sequence to sequence learning with neural networks.” NIPS 2014
- Poznanski, A., & Wolf, L. “CNN-N-gram for handwriting word recognition.” CVPR 2016
- Jaderberg, M., Simonyan, K., Vedaldi, A., & Zisserman, A. “Reading text in the wild with convolutional neural networks.” IJCV 2016
- Bojanowski, P., Grave, E., Joulin, A., & Mikolov, T. “Enriching word vectors with sub-word information.” TACL 2017

A.13 Style Transfer

Theory:

Style transfer aims to render content from one image in the style of another. For example: apply Van Gogh’s brushstrokes to a photograph.

Approaches:

- **Optimization-based:** Iteratively optimize pixels to minimize content and style losses computed via a pretrained CNN
- **Feed-forward networks:** Train a network to directly transform images; fast but less flexible
- **Instance normalization:** Align feature statistics between content and style
- **Adversarial / CycleGAN:** Unsupervised learning without style pairs
- **Text-guided:** Control style with natural language (e.g., “oil painting”)

Key insight: different CNN layers capture different information; deep layers capture style (colors, textures), while intermediate layers capture content (objects, layout).

Recent work uses diffusion models and large language models for fine-grained control.

Key Papers:

- Hertzmann, A., Jacobs, C. E., Oliver, N., Curless, B., & Salesin, D. H. “Image analogies.” SIGGRAPH 2001
- Gatys, L. A., Ecker, A. S., & Bethge, M. “Image style transfer using convolutional neural networks.” CVPR 2016
- Li, Y., Fang, C., Yang, J., Wang, Z., Lu, X., & Yang, M. H. “Universal Style Transfer via Feature Transforms.” arXiv 2017
- Huang, X., & Belongie, S. “Arbitrary style transfer in real-time with adaptive instance normalization.” CVPR 2017
- Huang, X., Liu, M. Y., Belongie, S., & Kautz, J. “Multimodal unsupervised image-to-image translation.” ECCV 2018
- **BONUS:** Kwon, G., & Ye, J. C. “ClipStyler: Image Style Transfer with a Single Text Condition.” CVPR 2022

A.14 Disentangling Representations

Theory:

Disentangled representations separate different “factors of variation” into independent dimensions. For example, for faces: pose, lighting, identity, expression should each be represented separately.

Why it matters:

- Interpretability: dimensions have clear meaning
- Compositionality: combine factors in novel ways
- Robustness: factors are independent, easier to learn
- Few-shot learning: change one factor while others remain fixed

Approaches:

- Variational Autoencoders (VAEs): impose independence via prior
- InfoGAN: maximize mutual information between codes and observations
- Slot Attention: assign image regions to “slots” (objects)

- Weakly-supervised: use auxiliary information (e.g., transformation labels)

Challenges: defining what constitutes a “factor”; proving disentanglement theoretically.

Warning: Rather difficult topic; foundational concepts matter. Focus on core papers first.

Key Papers:

- Cheung, B., Livezey, J. A., Bansal, A. K., & Olshausen, B. A. “Discovering hidden factors of variation in deep networks.” arXiv 2014
- Kulkarni, T. D., Whitney, W. F., Kohli, P., & Tenenbaum, J. B. “Deep convolutional inverse graphics network.” NIPS 2015
- Yang, J., Reed, S. E., Yang, M. H., & Lee, H. “Weakly-supervised disentangling with recurrent transformations for 3D view synthesis.” NIPS 2015
- Chen, X., Duan, Y., Houthoofd, R., Schulman, J., Sutskever, I., & Abbeel, P. “InfoGAN: Interpretable representation learning by information maximizing generative adversarial nets.” NIPS 2016
- Shu, Z., Yumer, E., Hadap, S., Sunkavalli, K., Shechtman, E., & Samaras, D. “Neural face editing with intrinsic image disentangling.” CVPR 2017
- Shu, Z., Sahasrabudhe, M., Guler, R. A., Samaras, D., Paragios, N., & Kokkinos, I. “Deforming autoencoders: Unsupervised disentangling of shape and appearance.” ECCV 2018
- Locatello, F., Weissenborn, D., Unterthiner, T., et al. “Object-centric learning with slot attention.” NeurIPS 2020

A.15 Human Shape and Pose Recognition and Estimation

Theory:

Human pose estimation answers: “What is the position and orientation of each body part?” This has applications in sports analytics, motion capture, healthcare (rehabilitation), and human-computer interaction.

Representations:

- **Keypoints:** Locations of joints (2D pixel coordinates or 3D world coordinates)
- **Skeletons:** Connected joints forming a kinematic tree
- **SMPL models:** Parametric 3D body shape and pose
- **Heatmaps:** Per-pixel confidence for each joint

Challenges:

- Occlusion (limbs hidden behind body or other objects)
- Perspective and foreshortening
- High-degree-of-freedom problem (many valid configurations)
- Generalizing across people, clothing, poses

Modern methods: CNN heatmap regression, graph neural networks (treating skeleton as graph), transformers, synthetic training data.

Key Papers:

- Shotton, J., Fitzgibbon, A., Cook, M., Sharp, T., Finocchio, M., Moore, R., Kipman, A., & Blake, A. “Real-time human pose recognition in parts from single depth images.” CVPR 2011
- Varol, G., Romero, J., Martin, X., Mahmood, N., Black, M. J., Laptev, I., & Schmid, C. “Learning from synthetic humans.” CVPR 2017
- Güler, R. A., Neverova, N., & Kokkinos, I. “DensePose: Dense human pose estimation in the wild.” CVPR 2018
- Kanazawa, A., Black, M. J., Jacobs, D. W., & Malik, J. “End-to-end recovery of human shape and pose.” CVPR 2018
- Carreira, J., Agrawal, P., Fragkiadaki, K., & Malik, J. “Human pose estimation with iterative error feedback.” CVPR 2016
- **BONUS:** Khirodkar, R., et al. “Sapiens: Foundation for Human vision models.” ECCV 2024

A.16 Domain Adaptation

Theory:

Domain adaptation addresses the shift between training data (source domain) and test data (target domain). For example: training on synthetic data but deploying on real images.

Challenges:

- Distribution mismatch: source and target have different feature distributions
- Label shift: class frequencies differ between domains
- Unlabeled target data: must adapt without target labels

Approaches:

- **Adversarial:** Fool a domain discriminator into treating source and target similarly
- **Correlation Alignment (CORAL):** Align feature covariances
- **Cycle Consistency:** Transform images back-and-forth while preserving content
- **Self-training:** Use predictions on target data as pseudo-labels

Applications: sim-to-real robotics, cross-camera surveillance, seasonal changes.

Key Papers:

- Csurka, C. “Domain Adaptation for Visual Applications: A Comprehensive Survey.” arXiv 2017 (*main overview; focus on key approaches*)
- Ganin, Y., & Lempitsky, V. “Unsupervised domain adaptation by backpropagation.” arXiv 2014
- Sun, B., Feng, J., & Saenko, K. “Return of frustratingly easy domain adaptation.” AAAI 2016
- Sun, B., & Saenko, K. “Deep CORAL: Correlation alignment for deep domain adaptation.” ECCV 2016
- Hoffman, J., Tzeng, E., Park, T., Zhu, J. Y., Isola, P., Saenko, K., et al. “CyCADA: Cycle-consistent adversarial domain adaptation.” arXiv 2017
- Tzeng, E., Hoffman, J., Saenko, K., & Darrell, T. “Adversarial discriminative domain adaptation.” CVPR 2017

A.17 Multi-Task Deep Learning

Theory:

Multi-task learning trains a single model to solve multiple related tasks simultaneously. Sharing representations can improve generalization and reduce overfitting.

Benefits:

- Shared representations capture common low-level patterns
- Regularization effect: tasks constrain each other
- Improved data efficiency (more training signal)

Challenges:

- Task interference: tasks compete for model capacity
- Balancing losses: different tasks may have different scales
- Negative transfer: one task hurts another's performance

Architectures:

- **Hard parameter sharing:** Common trunk network, task-specific heads
- **Soft parameter sharing:** Task-specific networks with regularization encouraging similarity
- **Cross-stitch networks:** Learned mixing of intermediate representations
- **Multi-objective optimization:** Pareto-optimal trade-offs between tasks

Key Papers:

- Misra, I., Shrivastava, A., Gupta, A., & Hebert, M. "Cross-stitch networks for multi-task learning." CVPR 2016
- Zamir, A. R., Sax, A., Shen, W., Guibas, L. J., Malik, J., & Savarese, S. "Taskonomy: Disentangling task transfer learning." CVPR 2018
- Kokkinos, I. "UberNet: Training a universal convolutional neural network for low-, mid-, and high-level vision using diverse datasets and limited memory." CVPR 2017
- Sener, O., & Koltun, V. "Multi-task learning as multi-objective optimization." NeurIPS 2018
- Eigen, D., & Fergus, R. "Predicting depth, surface normals and semantic labels with a common multi-scale convolutional architecture." ICCV 2015 (*focus on multi-task aspects*)

A.18 Deep Depth Prediction / 3D Reconstruction

Theory:

Given a single image (or video), can we estimate depth (distance of each pixel from camera)? This enables 3D understanding and reconstruction.

Approaches:

- **Supervised:** Train on ground-truth depth from sensors (Kinect, LiDAR)
- **Unsupervised:** Use photometric consistency across video frames; enforce geometric constraints

- **Self-supervised:** Combine depth and ego-motion estimation

Key ideas:

- Photometric loss: reprojected pixels should match original
- Geometric consistency: multiple views should agree
- Occlusion handling: visible pixels change across views

Applications: 3D reconstruction, autonomous driving (obstacle detection), augmented reality, robotics.

Key Papers:

- Eigen, D., & Fergus, R. “Predicting depth, surface normals and semantic labels with a common multi-scale convolutional architecture.” ICCV 2015
- Godard, C., Mac Aodha, O., & Brostow, G. J. “Unsupervised monocular depth estimation with left-right consistency.” CVPR 2017
- Zhan, H., Garg, R., Weerasekera, C. S., Li, K., Agarwal, H., & Reid, I. “Unsupervised learning of monocular depth estimation and visual odometry with deep feature reconstruction.” CVPR 2018
- Li, Z., & Snavely, N. “MegaDepth: Learning single-view depth prediction from internet photos.” CVPR 2018
- Vijayanarasimhan, S., Ricco, S., Schmid, C., Sukthankar, R., & Fragkiadaki, K. “SfM-Net: Learning of structure and motion from video.” arXiv 2017
- Tulsiani, S., Zhou, T., Efros, A. A., & Malik, J. “Multi-view supervision for single-view reconstruction via differentiable ray consistency.” CVPR 2017

A.19 Deep 3D Recognition

Theory:

3D recognition: classifying and analyzing objects represented as 3D point clouds or voxels (volumetric grids). This is crucial for robotics, autonomous driving, and 3D scene understanding.

Representations:

- **Point clouds:** Unordered sets of 3D coordinates; permutation-invariant
- **Meshes:** Surfaces with connectivity information
- **Voxels:** Regular 3D grids (like 3D pixels)
- **Implicit functions:** Continuous functions $\mathbb{R}^3 \rightarrow \mathbb{R}$

Key challenge: 3D data is high-dimensional and sparse. Standard CNNs don’t apply directly to point clouds.

PointNet innovations:

- Process point clouds directly without conversion
- Permutation-invariant via symmetric functions (max pooling)
- Learn both point-level and global features

Recent work: hierarchical processing (PointNet++), graph neural networks, transformers on 3D data.

Key Papers:

- Qi, C. R., Su, H., Mo, K., & Guibas, L. J. “PointNet: Deep learning on point sets for 3D classification and segmentation.” CVPR 2017
- Qi, C. R., Yi, L., Su, H., & Guibas, L. J. “PointNet++: Deep hierarchical feature learning on point sets in a metric space.” NeurIPS 2017
- Su, H., Jampani, V., Sun, D., Maji, S., Kalogerakis, E., Yang, M. H., & Kautz, J. “SplatNet: Sparse lattice networks for point cloud processing.” CVPR 2018
- Su, H., Maji, S., Kalogerakis, E., & Learned-Miller, E. “Multi-view convolutional neural networks for 3D shape recognition.” ICCV 2015
- Riegler, G., Osman Ulusoy, A., & Geiger, A. “Octnet: Learning deep 3D representations at high resolutions.” CVPR 2017
- Bronstein, M. M., Bruna, J., LeCun, Y., Szlam, A., & Vandergheynst, P. “Geometric deep learning: Going beyond Euclidean data.” IEEE Signal Processing Magazine 2017
- Landrieu, L., & Simonovsky, M. “Large-scale point cloud semantic segmentation with superpoint graphs.” CVPR 2018

A.20 Intrinsic Images

Theory:

An image is the product of shape (geometry), material properties (reflectance/albedo), and lighting. Intrinsic image decomposition aims to separate these factors:

$$\text{Image} = \text{Reflectance} \times \text{Shading}$$

Why it matters:

- Relighting: change lighting without reacquiring images
- Material recognition: identify surface properties
- Object recognition: reflectance more stable across lighting changes

Challenges:

- Ill-posed: many decompositions explain the same image
- Need strong assumptions (e.g., piecewise constant reflectance)
- Specularities and transparency complicate the model

Approaches:

- Optimization with priors (sparsity, smoothness)
- Learning from synthetic data
- Self-supervised using geometric constraints (shading must be smooth after deformation)

Key Papers:

- Barron, J. T., & Malik, J. “Shape, illumination, and reflectance from shading.” PAMI 2014
- Weiss, Y. “Deriving intrinsic images from image sequences.” ICCV 2001
- Bell, S., Bala, K., & Snavely, N. “Intrinsic images in the wild.” TOG 2014
- Zhou, T., Krahenbuhl, P., & Efros, A. A. “Learning data-driven reflectance priors for intrinsic image decomposition.” ICCV 2015
- Narihira, T., Maire, M., & Yu, S. X. “Direct intrinsics: Learning albedo-shading decomposition by convolutional regression.” ICCV 2015
- Li, Z., & Snavely, N. “Learning intrinsic image decomposition from watching the world.” CVPR 2018
- Janner, M., Wu, J., Kulkarni, T. D., Yildirim, I., & Tenenbaum, J. B. “Self-supervised intrinsic image decomposition.” NeurIPS 2017

A.21 Deep RANSAC

Theory:

RANSAC (RANDOM Sample Consensus) is a classical algorithm for robust fitting: given noisy data with outliers, fit a geometric model (line, homography, fundamental matrix) that ignores outliers.

Standard RANSAC:

1. Randomly sample minimal set (e.g., 2 points for a line)
2. Fit model
3. Count inliers
4. Repeat until confident model found

Limitations:

- Slow: many random samples needed for high outlier rates
- Hyperparameter-sensitive (inlier threshold)
- Doesn't learn from data

Deep learning approaches:

- Learn to find better correspondences (Neural Matching Networks)
- Learn to predict inlier probabilities
- Differentiable RANSAC: enable end-to-end optimization
- Attention mechanisms: focus on important correspondences

Applications: camera pose estimation, 3D reconstruction, image registration.

Key Papers:

- Yi, K. M., Trulls, E., Ono, Y., Lepetit, V., Salzmann, M., & Fua, P. “Learning to find good correspondences.” CVPR 2018
- Ranftl, R., & Koltun, V. “Deep fundamental matrix estimation.” ECCV 2018

- Li, L., Sung, M., Dubrovina, A., Yi, L., & Guibas, L. J. “Supervised fitting of geometric primitives to 3D point clouds.” CVPR 2019
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., et al. “Attention is all you need.” NeurIPS 2017
- Rocco, I., Cimpoi, M., Arandjelović, R., Torii, A., Pajdla, T., & Sivic, J. “Neighbourhood consensus networks.” NeurIPS 2018

A.22 Deep 3D Generation

Theory:

Generating 3D objects from 2D images or noise. Unlike 2D image generation, 3D generation must satisfy geometric constraints.

Representations:

- **Voxel grids:** Dense but computationally expensive
- **Point clouds:** Sparse and flexible
- **Meshes:** Structured surfaces, harder to learn
- **Implicit functions:** Continuous (neural fields), efficient rendering

Approaches:

- Shape primitives (cubes, spheres assembled into objects)
- Single-image 3D: encoder-decoder on learned representation
- Multi-view: assemble 2D predictions into consistent 3D
- Generative models: VAE or GAN in 3D

Key challenge: 3D ground truth is expensive; self-supervised learning from multi-view consistency.

Applications: object reconstruction, 3D asset creation, scene understanding.

Key Papers:

- Tulsiani, S., Su, H., Guibas, L. J., Efros, A. A., & Malik, J. “Learning shape abstractions by assembling volumetric primitives.” CVPR 2017
- Choy, C. B., Xu, D., Gwak, J., Chen, K., & Savarese, S. “3D-R2N2: A unified approach for single and multi-view 3D object reconstruction.” ECCV 2016
- Häne, C., Tulsiani, S., & Malik, J. “Hierarchical surface prediction for 3D object reconstruction.” 3DV 2017
- Tatarchenko, M., Dosovitskiy, A., & Brox, T. “Octree generating networks: Efficient convolutional architectures for high-resolution 3D outputs.” ICCV 2017
- Fan, H., Su, H., & Guibas, L. J. “A point set generation network for 3D object reconstruction from a single image.” CVPR 2017
- Groueix, T., Fisher, M., Kim, V. G., Russell, B. C., & Aubry, M. “AtlasNet: A Papier-Mâché Approach to Learning 3D Surface Generation.” CVPR 2018
- Chen, Z., & Zhang, H. “Learning implicit fields for generative shape modeling.” CVPR 2019

- Park, J. J., Florence, P., Straub, J., Newcombe, R., & Lovegrove, S. “DeepSDF: Learning continuous signed distance functions for shape representation.” CVPR 2019
- Mescheder, L., Oechsle, M., Niemeyer, M., Nowozin, S., & Geiger, A. “Occupancy networks: Learning 3D reconstruction in function space.” CVPR 2019
- Richter, S. R., & Roth, S. “Matryoshka networks: Predicting 3D geometry via nested shape layers.” CVPR 2018
- Sun, X., Wu, J., Zhang, X., Zhang, Z., Zhang, C., Xue, T., et al. “Pix3D: Dataset and methods for single-image 3D shape modeling.” CVPR 2018
- Tatarchenko, M., Richter, S. R., Ranftl, R., Li, Z., Koltun, V., & Brox, T. “What do single-view 3D reconstruction networks learn?” CVPR 2019

A.23 Cycle Consistency Supervision

Theory:

Cycle consistency: if you transform data in one direction and then back, you should recover the original. This provides self-supervision without paired data.

Key application: unpaired image-to-image translation (CycleGAN).

Examples:

- Image translation: horses $\leftrightarrow$ zebras (no paired training examples)
- Temporal correspondence: if point A matches B in frame t , then B should match C in frame $t + 1$ and C should match back to A eventually
- 3D geometry: dense correspondence in multiple views should be cycle-consistent

Why it works:

- Encodes geometric/semantic structure
- Reduces solution space dramatically
- Enables learning from unlabeled data

Applications: style transfer, unsupervised learning, correspondence estimation, domain adaptation.

Key Papers:

- Zhu, J. Y., Park, T., Isola, P., & Efros, A. A. “Unpaired image-to-image translation using cycle-consistent adversarial networks.” CVPR 2017
- Wang, X., Jabri, A., & Efros, A. A. “Learning correspondence from the cycle-consistency of time.” CVPR 2019
- Zhou, T., Krahenbuhl, P., Aubry, M., Huang, Q., & Efros, A. A. “Learning dense correspondence via 3D-guided cycle consistency.” CVPR 2016

A.24 Spectral Shape Analysis

Theory:

Spectral methods analyze 3D shapes using eigendecompositions of the Laplacian operator on the shape surface. This provides a coordinate-free, intrinsic representation of shape.

Key ideas:

- **Laplace-Beltrami operator:** Generalization of Laplacian to curved surfaces
- **Eigenfunctions:** Smooth basis functions; lower frequencies capture coarse geometry
- **Heat kernel:** Provides local and multi-scale information
- **Functional maps:** Represent correspondences between shapes as linear maps of functions

Advantages:

- Invariant to isometry (bending without stretching)
- Stable and interpretable
- Handles non-rigid shapes

Applications: shape matching, correspondence estimation, shape clustering, non-rigid registration.

This is a more specialized topic; good for students interested in geometry and mathematical foundations.

Key Papers:

- Sun, J., Ovsjanikov, M., & Guibas, L. “A concise and provably informative multi-scale signature based on heat diffusion.” CGF 2009
- Ovsjanikov, M., Ben-Chen, M., Solomon, J., Butscher, A., & Guibas, L. “Functional maps: A flexible representation of maps between shapes.” TOG 2012
- Litany, O., Remez, T., Rodola, E., Bronstein, A., & Bronstein, M. “Deep functional maps: Structured prediction for dense shape correspondence.” ICCV 2017

A.25 Class Activation Maps and Adversarial Attacks

Theory:

Understanding and attacking neural networks:

Class Activation Maps (CAMs):

- Visualize which regions of an image a CNN focuses on for classification
- Highlight discriminative regions without additional supervision
- Enable interpretability: does the model look at the object or background?

Adversarial Attacks:

- Tiny, imperceptible perturbations that fool classifiers
- Reveal vulnerabilities and lack of robustness
- Motivate better training techniques

Key findings:

- Models are surprisingly brittle to small adversarial perturbations
- Adversarial examples transfer across models (weak transferability)
- Can defend via adversarial training

Implications:

- Model interpretability is limited
- Deployment in safety-critical systems requires caution
- Adversarial robustness is an active research area

Key References:

- See: <https://arxiv.org/abs/1607.02533>
- See: <http://cnnlocalization.csail.mit.edu/>